Enhancing Learning Experiences For Visually Impaired Students With AI and Machine Learning on Smart Device

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2024 International Congress on Human-Computer Interaction, Optimization and Robotic Applications (HORA) | 979-8-3503-9463-4/24/$31.00 ©2024 IEEE | DOI: 10.1109/HORA61326.2024.10550610

***Abstract*—This paper explores the challenges faced by visually impaired students in educational settings. The research introduces a smart glass system based on YOLOv8 and Internet of Things technology, leveraging advancements in computer vision and machine learning. This system offers real-time object detection and comprehensive voice feedback, facilitating independent learning in subjects such as science, math, and financial transactions. The effectiveness of YOLOv8 has been proven through extensive training and analysis using a comprehensive dataset. By integrating text-to-speech and speech-to-text technologies, the smart glass system is reshaping inclusive education and underscores the importance of equal learning opportunities for visually impaired individuals worldwide, promoting independence, removing educational barriers, and fostering interest in STEM subjects among visually impaired students.**

Keywords—visually impaired students, artificial intelligence, machine learning, YOLO, IoT, education, object detection

I. INTRODUCTION

People use their eyes for nearly every activity and study. However, some individuals are unfortunately deprived of this ability. The World Health Organization estimates that there are currently 284 million visually impaired people worldwide, including 39 million who are blind [1]. In Sri Lanka, about 1.7% of the population is blind, approximately 364,000 people [2]. A survey revealed that among the study population, 1.6% had severe visual impairment and 15.4% had some form of visual impairment [3]. The Ceylon School for the Deaf and Blind, established in 1912, marked the beginning of education for the blind in Sri Lanka [2]. Despite an overall literacy rate of 92.5%, Sri Lanka, as a developing country, faces challenges with Braille literacy. According to the Ministry of Social Welfare, 71% of people who are blind or visually impaired have attended school at some level, yet only 6% have advanced beyond the G.C.E. Ordinary Level [4]. Therefore, it is crucial to enhance the educational system for visually impaired students by making it more accessible and engaging, thus increasing their interest in learning.

The research problem addressed in this context is the development of assistive technologies to enhance the education of visually impaired students. These students face challenges in science, math education, and object recognition. The aim is to leverage advancements in computer vision, machine learning, and accessibility technologies to create innovative solutions that improve accessibility for blind individuals. Currently, the favourite learning tools among visually impaired students include embossed images, real objects, and braille books. Traditional methods, which are time-consuming and often fail to engage students, result in low interest in learning. However, the time required to get acquainted with new technology can be reduced, potentially increasing learning interest among visually impaired students. It is generally more difficult for visually impaired students to learn than for their sighted peers. According to teachers at schools for the deaf and blind, learning science and mathematics poses additional challenges. Enhancing memory retention and sustaining attention among visually impaired students can be more effectively achieved through engaging activities rather than traditional methods. These activities not only strengthen memory but also maintain focus throughout the learning process. However, it is vital to acknowledge potential risks, particularly concerning the tactile learning of geometric and scientific objects, which may pose injury hazards. Additionally, the discomfort that visually impaired students experience when using money highlights the necessity for early financial literacy training. Such programs equip students with essential skills and confidence, facilitating smoother financial transactions.

Recognizing the significance of technology in education, a smart educational app paired with an Internet of Things (IoT) device proves invaluable for supporting the learning of visually impaired students. Object detection, combined with a user-friendly mobile application, offers an effective solution for enhancing learning activities and outcomes for these students.

II. LITERATURE REVIEW

Research into learning systems for the blind has captivated scholars for decades. According to a meaningful study, the ancient Egyptians were pioneers in addressing issues concerning visually impaired learners [1]. Every advancement in this field owes its existence to the meticulous research conducted by experts. Research continues, with new technologies being integrated into this domain each day. Lowenfeld emphasized the importance of educating blind children to help them understand the world, gain confidence, and feel acknowledged and embraced as unique individuals [2].

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Significant strides in assistive technology have profoundly impacted our scientific understanding of object detection for blind students. Initially, efforts to convey information about objects primarily focused on touch and auditory modalities. Tactile aids such as raised-line drawings and tactile graphics were developed to help visually impaired individuals understand shapes and diagrams. Over time, these aids evolved to include 3D models and embossed pictures, offering a more tangible representation of objects [3].

Although 71% of blind children are schooled, only 6% progress beyond the G.C.E. Ordinary Level [4]. Scoring in subjects like science and math is particularly challenging, with only 29% and 43% respectively [5]. Enhancing the desire to learn mathematics in blind students presents a formidable challenge. Blind students typically recognize money by touch and use a special braille system. They tend to learn better through hands-on activities rather than note-taking.

Historically, object detection for the blind relied on basic image processing techniques and rule-based algorithms. However, the integration of Convolutional Neural Networks (CNNs) has significantly improved the accuracy and efficiency of these systems. CNNs excel in image detection by autonomously learning features from raw data [1]. Additionally, the advent of You Only Look Once (YOLO), a real-time object detection system, has further enhanced the effectiveness of CNNs in object recognition for the visually impaired. Renowned for its speed and precision, YOLO is particularly well-suited for real-time applications that require rapid and accurate object recognition.

One of the key gaps highlighted is the lack of infrastructure and financial resources in schools, which hampers the proper implementation of technology. This issue is compounded by inadequate training for teachers and staff, limiting effective technology integration into the curriculum and ultimately restricting blind students' access to essential information and knowledge [4]. Despite technological advances, findings often lack generalizability, contain methodological flaws, and exhibit biases, which challenge their credibility [5]. Addressing these challenges requires more comprehensive research, as well as increased collaboration among researchers, educators, and technology developers to bridge these gaps and ensure equitable access to science education for blind students.

Interactive math study systems for blind students are innovative, yet many systems are limited to currency recognition, which restricts educational access. The Sri Lankan Currency Recognition Device for Visually Impaired People demonstrates progress in the field of currency identification. However, it lacks real-time feedback systems, mobile applications, and interaction with IoT devices, which diminishes its effectiveness in creating a welcoming and stimulating learning environment [6]. Singh's investigation into mobile phone money recognition introduced a mobile application for blind users [2]. While this is a positive development, the study focused solely on currency identification and overlooked other crucial aspects of mathematics instruction. Without mechanisms for real-time feedback, visually impaired students may face greater challenges in their studies [7].

A comprehensive study introduced a system capable of recognizing Euro banknotes and providing vocal feedback, significantly enhancing the ability of blind individuals to identify currency. Nevertheless, this study did not leverage an IoT-based device and could not identify multiple banknotes simultaneously. Its potential impact on mathematics education was limited due to the absence of robust integration with real-world arithmetic problems [8].

Despite improvements in assistive technology for students with visual impairments, many challenges remain. Traditional tactile aids, while helpful, may cover only a limited scope. For instance, raised-line drawings can be challenging to create and interpret, and physical models may not faithfully represent complex geometric designs.

Moreover, existing digital solutions often fall short in providing comprehensive access to mathematical content. Screen readers and text-to-speech software might struggle with accurately comprehending mathematical notation, which is essential for accessing digital text. This limitation can make it more difficult for students who are visually challenged to interact with mathematical concepts, potentially leading to frustration and disengagement.

One of the most significant challenges faced by blind students is the time-consuming nature of the learning process. In the absence of readily available educational resources and effective teaching strategies, students with visual impairments often require more time to fully grasp mathematical content. This discrepancy in learning time can hinder academic development and contribute to feelings of isolation.

III. Methodology

A. Theoretical Framework

This section provides a comprehensive overview of the theories that have been employed throughout this research and highlights the significance and relevance of each theory.

In the realm of object identification models, three algorithms stand out: Mask R-CNN, YOLO, and Faster R-CNN. Each model has distinct features and suits a range of applications, depending on its benefits and weaknesses. YOLO, known for its real-time object detection capabilities, divides the input image into a grid of cells and predicts the bounding boxes and class probabilities for each cell simultaneously. Although YOLO sacrifices some accuracy for speed, it remains the preferred choice for applications requiring fast processing. In contrast, Faster R-CNN achieves a balance between speed and accuracy by employing a two-stage detection technique. It first generates region proposals using a Region Proposal Network (RPN), then refines these proposals with a Fast R-CNN network. This method excels when accuracy is more critical than real-time speed in object detection scenarios. Mask R-CNN extends the capabilities of Faster R-CNN by adding a segmentation mask prediction branch, offering pixel-level precision. It performs exceptionally well in tasks requiring precise instance segmentation, despite its substantial computational demand, making it suitable for applications that demand an in-depth understanding of object boundaries.

As shown in Fig. 1, the YOLO algorithm processes the input image to generate a grid of cells. Within each cell, YOLO simultaneously predicts bounding boxes and class

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probabilities. Each bounding box includes class probabilities for various item categories and a confidence score, which indicates the likelihood of an object being present in the predicted box. This prediction process utilizes a single convolutional neural network (CNN), which has been trained end-to-end to optimize both localization and classification tasks. During inference, the YOLO model predicts bounding boxes and class probabilities for each grid cell. These predictions are then refined through post-processing techniques such as non-maximum suppression (NMS). NMS enhances detection accuracy by eliminating redundant bounding boxes, selecting the one with the highest confidence score among those that overlap.

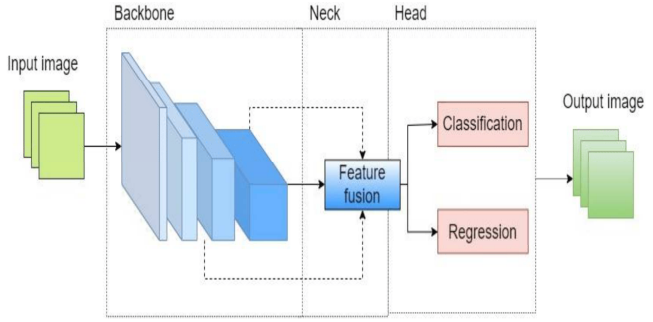


Fig. 1. YOLO Architecture

When it comes to development environments, Google Colab and Jupyter Notebook each offer distinct advantages. Google Colab is a cloud-based platform that provides free access to GPU and TPU resources, thus eliminating the need for powerful local hardware and enabling advanced deep learning research. Its collaboration features allow multiple users to work simultaneously on the same notebook, and its integration with Google Drive simplifies access to datasets stored in the cloud. On the other hand, Jupyter Notebook is an open-source web application that allows users to create and share documents containing live code, equations, visualizations, and narrative text. It can run locally on a user's computer, providing offline access and greater control over computing resources, making it the preferred choice for users who value customization and flexibility.

The functions and applications of Raspberry Pi and Arduino boards are distinctly different. The Raspberry Pi functions as a single-board computer and is ideal for a variety of applications requiring substantial computing power, such as media centers and IoT applications. It boasts a robust processor, ample RAM, and a variety of networking options. In contrast, the Arduino's microcontroller-based design, which offers simplicity and real-time control, makes it excellent for embedded systems and electronics projects involving sensors, actuaries, and automation. Although the Raspberry Pi presents a steeper learning curve than the Arduino due to its more complex architecture and operating system requirements, it also offers greater flexibility and a broader range of capabilities, albeit at a higher cost.

As illustrated in Fig. 2, a mobile application was developed using Flutter for the client-side interface, while the server-side logic was implemented using Python. WebSocket technology was utilized to enable real-time, bidirectional, full-duplex communication over a single TCP connection between the Flutter client and the Python server.

Flutter enabled cross-platform mobile app development, ensuring compatibility and a consistent user experience on both iOS and Android devices with a single codebase. Its extensive widget library, expressive UI components, and hot reload feature expedited development and facilitated rapid UI prototyping. Python was chosen for the server-side due to its versatility, robustness, and comprehensive library support. WebSocket communication was efficiently implemented using Python's asyncio package, which allows the server to manage multiple client connections concurrently while maintaining responsiveness.

B. Proposed system

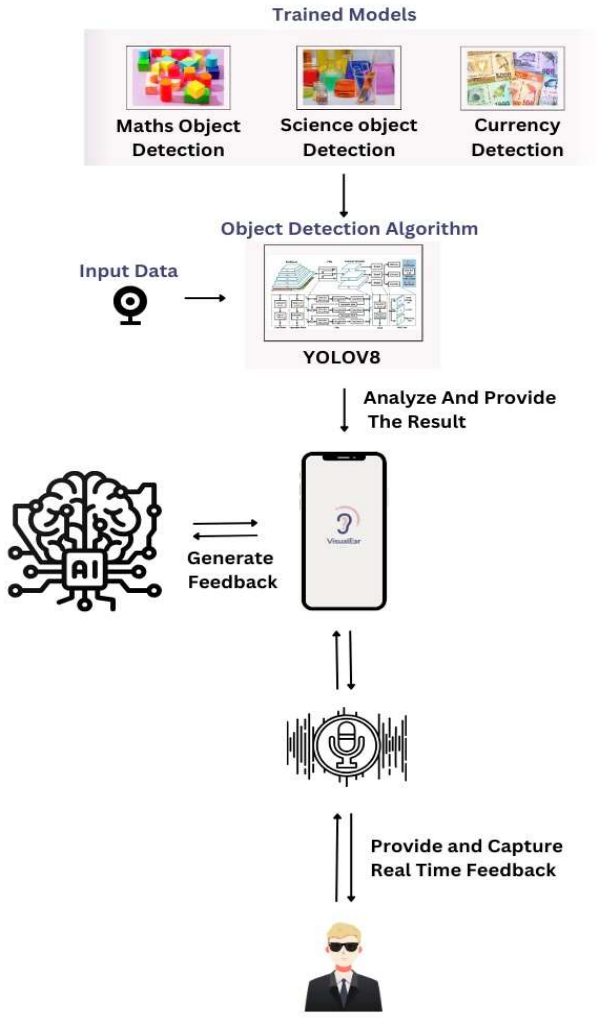


Fig. 2. System diagram

In this system, all functions utilized the YOLO algorithm, implemented using the OpenCV library in Python. OpenCV provides robust computer vision capabilities, enabling efficient object detection tasks. NumPy was also utilized for mathematical calculations and array manipulation, which simplified the pre-processing and post-processing stages of the object recognition pipeline. The geometric shape detection capabilities were seamlessly integrated with accessibility features to support visually challenged students. The system uses the Flutter Text-to-Speech (TTS) library to provide audio feedback, enabling blind users to hear descriptions of recognized geometric shapes in real-time. Moreover, for every detected shape, dynamically produced descriptive text is generated by OpenAI GPT-3.5. The science equipment identification function incorporates several significant technologies to enhance accessibility and provide visually impaired students with an extensive

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educational experience. This feature uses the Flutter TTS toolkit to produce audio outputs, allowing blind students to receive real-time audio descriptions of equipment in science labs.

Furthermore, arbitrary descriptions for these items have been generated using OpenAI GPT-3.5, enriching the educational content with dynamic and contextually relevant information. Additionally, blind students can guess the equipment name by speaking into the system, with the Flutter Speech-to-Text library capturing their guesses. Gen AI is utilized to create random questions for the currency detection feature, while Flutter TTS and Speech-to-Text facilitate audio outputs and user interaction. Thanks to Gen AI's advanced natural language generation capabilities, a wide range of contextually relevant questions regarding money and transactions for visually impaired students can now be constructed in real time. This system uses Gen AI to provide visually impaired users with engaging and instructional content, enhancing their educational experiences, engagement, and understanding. As depicted in Fig. 3, instance segmentation is utilized to accurately capture the responses of blind students using currency notes and coins, ensuring precise interpretation and interaction within the educational context.



Fig. 3. Instance Segmentation

1) User Interaction Flow:

The user interaction flow of the system encompasses three key functionalities: Geometric Shape Detection Model (GSDM), Scientific Equipment Detection Model (SEDM), and Currency Recognition Model (CRM), each designed to cater to the unique educational needs of blind and visually impaired students.

Students can quickly navigate between these elements from the main menu upon launching the mobile application. In the Geometric Shape Detection, students select geometric objects to examine and capture images with camera-equipped smart glasses. The system uses the YOLO object detection paradigm for real-time detections and provides auditory feedback including the name, characteristics, and relevant mathematical equations for calculating parameters such as area, perimeter, and volume. This interactive approach promotes independent learning and comprehension of geometric concepts.

Similarly, students can access the Laboratory Equipment Learning from the application's home page. Once there, users can guess the name of the object by interacting with its interfaces and engaging with science laboratory equipment. Upon accurately guessing an object's name, the system emits an alert stating, "You are correct." If the guessed name is incorrect, the system provides audio feedback, "You are wrong," followed by the correct name and a random description of the particular object.

The CRM enables students to solve mathematical exchange problems from real-world scenarios. They calculate answers and visually demonstrate their responses using physical currency. Employing YOLO object detection, the system verifies the accuracy of calculations and provides real-time audio feedback. If the displayed answer is correct, the system displays "Your answer is correct"; if not, it specifies what the mistake is.

2) Types of Data Used and Data Collection Methods:

Images of geometric forms commonly used to teach mathematics to blind or visually impaired students are included in the dataset used for this study. These images were captured at the Ratmalana Blind School, specifically in classrooms housing fifth and sixth-grade students. To ensure the completeness of the dataset, photographs of various geometric forms, tactile aids, educational materials, science laboratory equipment, and coins and currency notes used during instruction were taken.

Data collection efforts involved direct engagement with the classroom environment, enabling the capture of real and representative resources used in science and math classes for students with vision impairments. Additionally, photos of coins and currency notes were gathered from daily transactions and trips to nearby banks, ensuring a wide variety of denominations and variants commonly encountered in everyday life.

High-quality digital cameras were used to capture images of the objects, ensuring the images were suitable for subsequent analysis and annotation. The primary objective of this data collection approach was to obtain accurate and comprehensive representations of the geometric shapes, scientific lab equipment, and currency used in instructional methods as well as those encountered in real-world scenarios.

Subsequently, each image in the collection was meticulously annotated using LabelImg software. During the annotation process, coin borders and characteristics, banknotes, laboratory objects, and geometric shapes were all marked.

3) Tools and procedures for Dataset Creation:

To enhance the quality and diversity of the dataset, several preprocessing and augmentation techniques were applied after data collection. The labelImg tool facilitated the annotation process by accurately classifying the geometric shapes visible in the images. This meticulous annotation enabled the precise identification and naming of every geometric item in the dataset.

After annotation, the dataset underwent preprocessing using Roboflow, a powerful platform designed for data preprocessing and optimization. Roboflow standardized and optimized the dataset through various methods, including scaling, grayscale conversion, object isolation, and static cropping. These preprocessing techniques were essential for improving image resolution, standardizing image formats, and isolating geometric shapes for further analysis.

To increase the robustness and diversity of the dataset, augmentation techniques were also employed. These techniques, including flipping, rotating, blurring, and adding

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random noise, were used to simulate variations commonly observed in real-world scenarios, thus enhancing the model's generalization and performance.

IV. Results and Discussion

After evaluating different versions of YOLO, YOLOv8 was selected for its superior accuracy and faster training capabilities. Subsequently, YOLOv8 was used to train three object detection models: the CRM, Laboratory Equipment Detection Model (LEDM), and Math Object Detection Model (MODM). This approach provided a consistent architectural foundation across all models, facilitating efficient training and optimizing performance in their respective object detection tasks.

To enhance the models' generalization capabilities and improve object recognition under various lighting conditions and orientations, data augmentation techniques were implemented. These techniques included horizontal and vertical flipping of images, random rotations, and color jittering, which adjusted brightness, contrast, and saturation. Roboflow, a user-friendly data manipulation library, facilitated the efficient implementation of these augmentations on the dataset. A crucial step in the training process involved splitting the dataset into distinct sets for training, testing, and validation. A 70/20/10 split was adopted, allocating 70% of the data for model training, 20% for testing, and the remaining 10% for validation. This distribution helped monitor the model's learning progress and mitigated the risk of overfitting.

The training was conducted on the cloud-based platform Google Colab, which provided a convenient and accessible workspace. The availability of Tensor Processing Units (TPUs) in Colab significantly reduced training time and expedited results. TPUs are specifically designed for deep learning tasks.

To achieve an optimal balance between training efficiency and model performance, the impact of three key hyperparameters—batch size, image size, and epochs—was investigated. 'Batch size' refers to the number of images processed together during each training iteration, 'image size' determines the resolution of the images fed into the model, and 'epochs' represent the number of times the entire training dataset is passed through the model. Image sizes ranged from 480x480 to 1280x720. Lower resolutions (e.g., 170x170) might limit the model's ability to extract crucial features, while very high resolutions (e.g., 1280x720) could increase computational demands. Various batch sizes were explored, including 1, 8, 16, and 32, and different numbers of epochs, ranging from 10 to 50, were tested. Through this exploration, a configuration of a batch size of 16, an image size of 640x480, and 30 epochs was identified as delivering the best trade-off between training speed, model accuracy, and convergence. This configuration effectively leveraged computational resources while ensuring robust features were learned for accurate object detection. Figures 4, 5, and 6 visually represent the experimentation process and the optimal configuration.

To evaluate the effectiveness of the object detection models, confusion matrices were utilized, as shown in Table 1. These matrices provide a comprehensive view of the models' classification performance, detailing the number of true positives, true negatives, false positives, and false negatives for each object class the models are trained to detect. Fig. 7 and Fig. 8 present the confusion matrices for the CRM, LEDM, and the GSDM. By analyzing these matrices, valuable insights into the models' ability to correctly identify various objects within their respective domains can be gleaned. This information can guide further optimization efforts to enhance the models' performance in real-world applications.

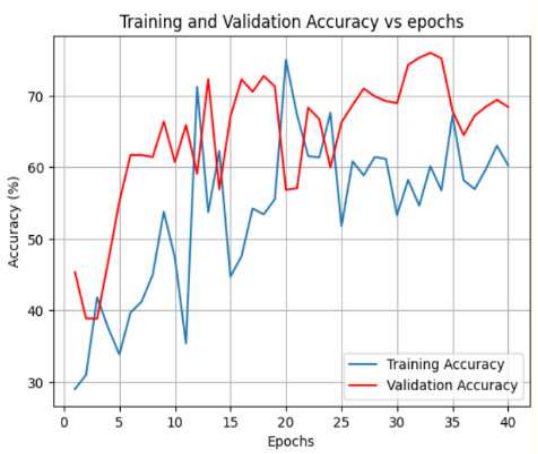


Fig. 4. The training and validation accuracy vs epochs chart of MODM

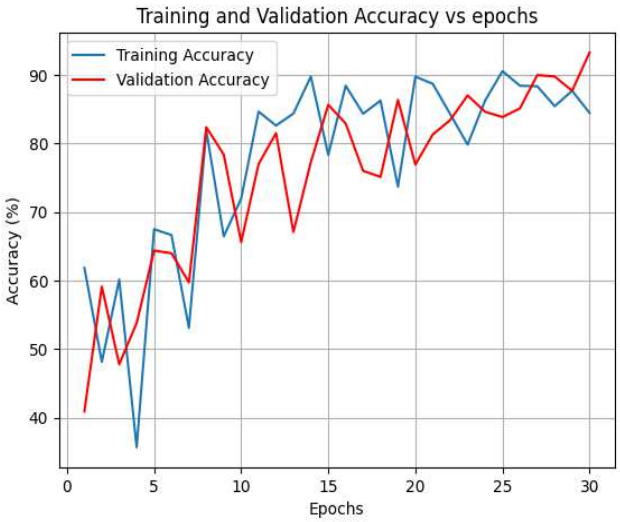


Fig. 5. The training and validation accuracy vs epochs chart of LEDM

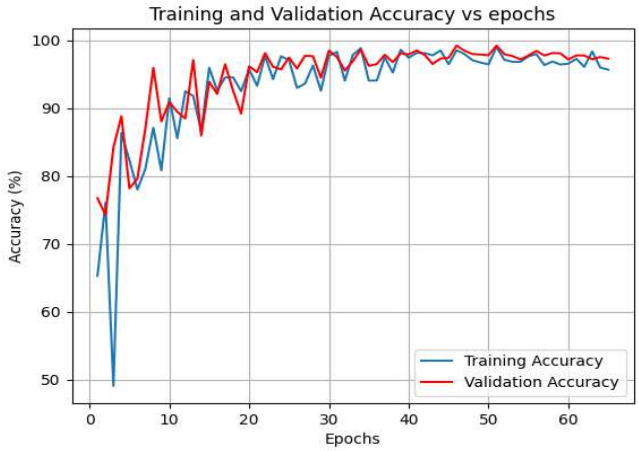


Fig. 6. The training and validation accuracy vs epochs chart of Sri Lankan Currency detection model

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TABLE I. TESTING ACCURACIES FOR EACH YOLO VERSION

YOLO version	SEDM		CRM		GSDM	
	Object detection	Instance segmentation	Object detection	Instance segmentation	Object detection	Instance segmentation
V5	61.32	-	72.00	-	70.99	-
V7	77.35	-	78.97	-	75.43	-
V8	88.67	76.00	83.57	80.32	85.78	81.33

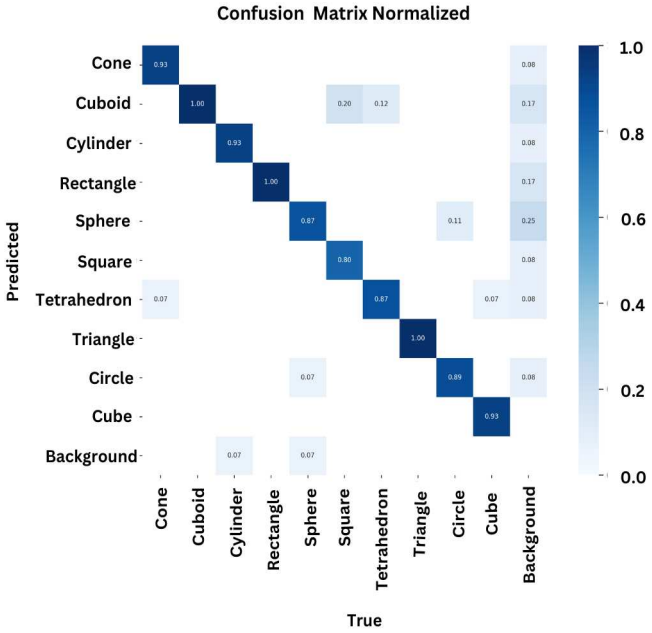


Fig. 6. Confusion matrix normalized chart of math object detection model

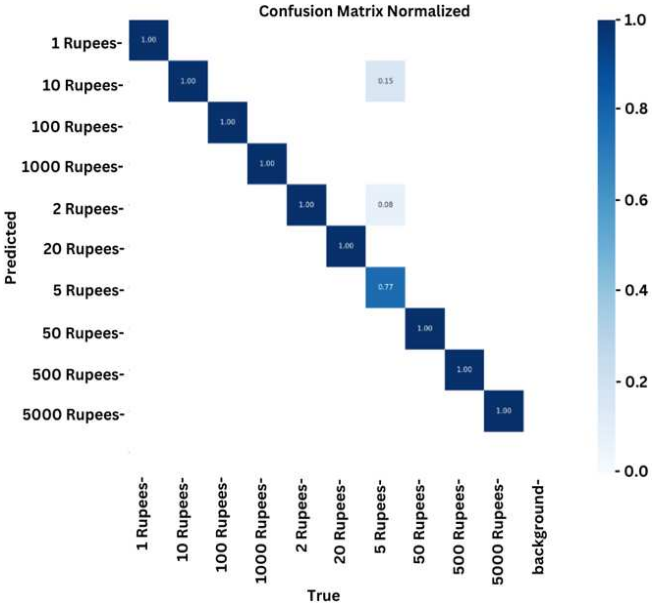


Fig. 7. Confusion matrix normalized chart of Sri Lankan currency detection model

CONCLUSION AND FUTURE WORK

This research has developed a smart glass system poised to revolutionize the education of blind people in science, mathematics, and currency usage. Blind students are empowered through the use of YOLOv8, speech-to-text, and text-to-speech technologies, facilitating independent learning, accurate object detection, and comprehensive descriptions. Potential improvements could include providing audio feedback in multiple languages, reducing the weight of the IoT device to ensure comfortable wear for blind students, and enabling standalone functionality that does not rely on a mobile application. This would allow the device to independently capture and provide voice commands. The findings are significant for educators and technologists working on inclusive education and accessibility. This study underscores the necessity of offering equal learning opportunities for all blind individuals, regardless of their visual ability. The impacts of this study extend beyond academia, influencing the daily lives and future aspirations of blind people worldwide. By aiming to promote inclusivity and equity, this research seeks to remove barriers to education and independence for blind students.

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